The SAT problem

Define

$$\mathbb{B} = \{0, 1\}$$

We want to answer the question, for a boolean function $f : \mathbb{B}^n \to \mathbb{B}$ which consists purely of the conjunction $\wedge$, disjunction $\vee$, and negation operator $\neg$; we want to determine:

$$\exists \mathbf{x} \in \mathbb{B}^n \text{ such that } f(\mathbf{x}) = 1$$

Motivating the SAT problem through reductions to SAT

Searching for tautologies

Suppose we want to determine whether the boolean function $g : \mathbb{B}^n \to \mathbb{B}$ is a tautology, that is, we want to determine whether:

$$\forall \mathbf{x} \in \mathbb{B}^n, g(\mathbf{x}) = 1$$

We want to make this in the form of the SAT problem.

To do this:

$$\neg \exists \mathbf{x} \in \mathbb{B}^n, g(\mathbf{x}) = 0$$

We define:

$$f(\mathbf{x}) = \neg g(\mathbf{x})$$

Thus, our problem becomes:

$$\neg \exists \mathbf{x} \in \mathbb{B}^n, f(\mathbf{x}) = 1$$

So we just solve the SAT problem and then negate the answer.

Determining equivalences

Suppose that we have two boolean functions $g : \mathbb{B}^n \to \mathbb{B}$ and $h : \mathbb{B}^n \to \mathbb{B}$.

The question is:

$$\forall \mathbf{x} \in \mathbb{B}^n, g(\mathbf{x}) == h(\mathbf{x})$$

This means that we can define:

$$f(\mathbf{x}) = \neg\left(g(\mathbf{x}) == h(\mathbf{x})\right)$$

And then ask the question:

$$\forall \mathbf{x} \in \mathbb{B}^n, f(\mathbf{x}) = 0$$

And so:

$$\neg\exists \mathbf{x} \in \mathbb{B}^n, f(\mathbf{x}) = 1$$

Note, we see that for 2 boolean variables $p, q \in \mathbb{B}$:

$$p = q \iff p \oplus q = 0$$

Therefore, we can even define:

$$f(\mathbf{x}) = g(\mathbf{x}) \oplus h(\mathbf{x})$$

Note that XOR operator $\oplus$ can be formed from just the $(\wedge, \vee, \neg)$ symbols, that is:

$$p \oplus q = (p \wedge \neg q) \vee (\neg p \wedge q)$$

We can place it in CNF form:

$$p \oplus q = (\neg(\neg p \vee q)) \vee (\neg(p \vee \neg q)) = (\neg p \vee q) \wedge (p \vee \neg q)$$

K-Coloring problem to SAT

Let's first define the K-coloring problem for graphs.

A graph can be defined by the set of edges and verticies.

That is, first, we have a set of vertices V, and then we have a set of edges E such that all elements in the set composes of another sets going from one vertex to another vertex in V.

The K-coloring first assigns k colors, say in the set C such that $|C| = k$.

Essentially, we want to determine a coloring function $c : V \to C$, such that:

$$\forall (i, j) \in V; c(i) \neq c(j)$$

To turn this into SAT, we create boolean variables:

$$C_{c,v}$$

Where we override the definition of c to be simply be an element:

$$c \in C$$

And:

$$v \in V$$

First of all, we encode the fact that for each $v \in V$, we only assign one color.

To do this, let's say that we have the one hot encoded function:

$$a(c, v) = \bigcap_{i=1}^{c-1} (c_{i,v}) \cap (\neg c_{c,v}) \cap \bigcap_{i=c+1}^{|c|} (c_{i,v})$$